Residual Dominance Drives Last-Item Reliance in Causal Self-Attention for Sequential Recommendation (Supplementary Document)

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1 Further Details on Norm-Based Analysis

In this appendix, we provide the full mathematical derivation of the norm-based analysis used in Section 4. We follow the framework of ?], who extended the norm-based analysis of ?] from multi-head attention alone to the entire attention block (multi-head attention, residual connection, and layer normalization). We describe how each component is decomposed and how the context-mixing ratio is computed at each stage.

1.1 Attention Block Formulation

We consider the attention block used in SASRec [?]. Given input representations $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_n] \in \mathbb{R}^{n \times d}$, the output at position i is:

$$\tilde{\mathbf{x}}_i = \mathrm{LN}(\mathrm{Attn}(\mathbf{x}_i, \mathbf{X}) + \mathbf{x}_i)\,, \tag{1}$$

where $\mathbf{x}_i \in \mathbb{R}^d$ denotes the i-th input representation (row vector), and $\tilde{\mathbf{x}}_i \in \mathbb{R}^d$ is the corresponding output. This block consists of three components: multi-head attention (Attn), residual connection, and layer normalization (LN).

Our goal is to decompose $\tilde{\mathbf{x}}_i$ into the sum of vector-valued functions of each input $\mathbf{x}_j$, and quantify their relative magnitudes.

1.2 Decomposition of Multi-Head Attention

The H-head attention output is:

$$\mathrm{Attn}(\mathbf{x}_i, \mathbf{X}) = \sum_{h=1}^{H} \mathrm{Attn}^{(h)}(\mathbf{x}_i, \mathbf{X}), \tag{2}$$

where each head computes:

$$\mathrm{Attn}^{(h)}(\mathbf{x}_i, \mathbf{X}) = \sum_{j=1}^{n} \alpha_{i,j}^{(h)} \left(\mathbf{x}_j \mathbf{W}_V^{(h)} + \mathbf{b}_V^{(h)} \right) \mathbf{W}_O^{(h)}, \tag{3}$$

with attention weights:

$$\alpha_{i,j}^{(h)} = \mathrm{softmax}_{\mathbf{x}_j \in \mathbf{X}} \left(\frac{(\mathbf{x}_i \mathbf{W}_Q^{(h)} + \mathbf{b}_Q^{(h)})(\mathbf{x}_j \mathbf{W}_K^{(h)} + \mathbf{b}_K^{(h)})^\top}{\sqrt{d_h}} \right). \tag{4}$$

Here, $\mathbf{W}_Q^{(h)}, \mathbf{W}_K^{(h)}, \mathbf{W}_V^{(h)} \in \mathbb{R}^{d \times d_h}$ and $\mathbf{W}_O^{(h)} \in \mathbb{R}^{d_h \times d}$ are weight matrices, and $\mathbf{b}_Q^{(h)}, \mathbf{b}_K^{(h)}, \mathbf{b}_V^{(h)} \in \mathbb{R}^{d_h}$ are bias vectors, with $d_h = d/H$.

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Conference'17, Washington, DC, USA

ACM ISBN 978-1-4503-XXXX-X/2026/06
https://doi.org/XXXXXXX.XXXXXXX

Since SASRec uses causal (left-to-right) self-attention, $\alpha_{i,j}^{(h)} = 0$ for $j > i$.

We define the value transformation for each head as:

$$f^{(h)}(\mathbf{x}) := \left(\mathbf{x} \mathbf{W}_V^{(h)} + \mathbf{b}_V^{(h)} \right) \mathbf{W}_O^{(h)}. \tag{5}$$

Substituting into (2) and (3), the attention output decomposes into a sum over input positions:

$$\mathrm{Attn}(\mathbf{x}_i, \mathbf{X}) = \sum_{j=1}^{n} \sum_{h=1}^{H} \alpha_{i,j}^{(h)} f^{(h)}(\mathbf{x}_j). \tag{6}$$

This decomposition, introduced by ?], shows that the attention output is naturally a sum of transformed input vectors weighted by attention.

1.3 Incorporating the Residual Connection

Adding the residual connection to (6):

$$\mathbf{y}_i := \mathrm{Attn}(\mathbf{x}_i, \mathbf{X}) + \mathbf{x}_i = \sum_{h=1}^{H} \sum_{j=1}^{n} \alpha_{i,j}^{(h)} f^{(h)}(\mathbf{x}_j) + \mathbf{x}_i. \tag{7}$$

Since the residual term $\mathbf{x}_i$ involves only position i, the additive structure is preserved. We separate the right-hand side into per-position contributions $\mathbf{y}_{i,j}$ as follows.

For $j \neq i$ (contributions from context):

$$\mathbf{y}_{i,j} := \sum_{h=1}^{H} \alpha_{i,j}^{(h)} f^{(h)}(\mathbf{x}_j). \tag{8}$$

For $j = i$ (contribution from self, via both attention and residual):

$$\mathbf{y}_{i,i} := \sum_{h=1}^{H} \alpha_{i,i}^{(h)} f^{(h)}(\mathbf{x}_i) + \mathbf{x}_i. \tag{9}$$

By construction, $\mathbf{y}_i = \sum_j \mathbf{y}_{i,j}$.

1.4 Decomposition of Layer Normalization

Layer normalization [?] is defined as:

$$\mathrm{LN}(\mathbf{y}) = \frac{\mathbf{y} - m(\mathbf{y})}{s(\mathbf{y})} \odot \boldsymbol{\gamma} + \boldsymbol{\beta}, \tag{10}$$

where $\odot$ denotes element-wise multiplication, and

$$m(\mathbf{y}) = \frac{1}{d} \sum_{k=1}^{d} y^{(k)}, \quad s(\mathbf{y}) = \sqrt{\frac{1}{d} \sum_{k=1}^{d} \left(y^{(k)} - m(\mathbf{y}) \right)^2 + \epsilon}, \tag{11}$$

with $y^{(k)}$ denoting the k-th element of $\mathbf{y}$, and $\epsilon > 0$ a small constant for numerical stability. Here, $\boldsymbol{\gamma}, \boldsymbol{\beta} \in \mathbb{R}^d$ are learnable parameters for element-wise scaling and shifting, respectively.

We now show that LN admits a "distributive law" over the additive decomposition of its input. Let $\mathbf{y}_i = \sum_j \mathbf{y}_{i,j}$ be the input to LN. Substituting into (10):

$$\mathrm{LN}(\mathbf{y}_i) = \frac{\mathbf{y}_i - m(\mathbf{y}_i)}{s(\mathbf{y}_i)} \odot \boldsymbol{\gamma} + \boldsymbol{\beta} = \frac{\sum_j \mathbf{y}_{i,j} - m\left(\sum_j \mathbf{y}_{i,j}\right)}{s(\mathbf{y}_i)} \odot \boldsymbol{\gamma} + \boldsymbol{\beta}. \quad (12)$$

The mean operator $m(\cdot)$ is a linear function (it computes the arithmetic mean of the vector elements). Therefore:

$$m\left(\sum_j \mathbf{y}_{i,j}\right) = \sum_j m(\mathbf{y}_{i,j}). \quad (13)$$

Substituting (13) into (12):

$$\begin{aligned} \mathrm{LN}(\mathbf{y}_i) &= \frac{\sum_j \mathbf{y}_{i,j} - \sum_j m(\mathbf{y}_{i,j})}{s(\mathbf{y}_i)} \odot \boldsymbol{\gamma} + \boldsymbol{\beta} && (14)\\ &= \frac{\sum_j \left(\mathbf{y}_{i,j} - m(\mathbf{y}_{i,j})\right)}{s(\mathbf{y}_i)} \odot \boldsymbol{\gamma} + \boldsymbol{\beta} && (15)\\ &= \sum_j \frac{\mathbf{y}_{i,j} - m(\mathbf{y}_{i,j})}{s(\mathbf{y}_i)} \odot \boldsymbol{\gamma} + \boldsymbol{\beta}. && (16) \end{aligned}$$

In the last step, we use the fact that the denominator $s(\mathbf{y}_i)$ is a scalar computed from the *entire* input $\mathbf{y}_i = \sum_j \mathbf{y}_{i,j}$, and is therefore a constant with respect to the summation over j. This allows it to be factored out of the sum. Note that, unlike $m(\cdot)$, the standard deviation $s(\cdot)$ is a nonlinear function and cannot be distributed over the sum. Instead, $s(\mathbf{y}_i)$ serves as a shared normalization factor across all components $\mathbf{y}_{i,j}$.

Defining:

$$g_{\mathbf{y}_i}(\mathbf{y}_{i,j}) := \frac{\mathbf{y}_{i,j} - m(\mathbf{y}_{i,j})}{s(\mathbf{y}_i)} \odot \boldsymbol{\gamma}, \quad (17)$$

we obtain the decomposition:

$$\tilde{\mathbf{x}}_i = \mathrm{LN}(\mathbf{y}_i) = \sum_j g_{\mathbf{y}_i}(\mathbf{y}_{i,j}) + \boldsymbol{\beta}. \quad (18)$$

This decomposition is *exact*; no approximation is involved. The numerator of each term $g_{\mathbf{y}_i}(\mathbf{y}_{i,j})$ is specific to the component $\mathbf{y}_{i,j}$ (the mean $m(\mathbf{y}_{i,j})$ is computed from $\mathbf{y}_{i,j}$ alone), while the denominator $s(\mathbf{y}_i)$ is shared across all components.

1.5 Full Decomposition

Combining the decompositions of Attn (§6), residual connection (§8–9), and LN (§18), the output of the attention block is:

$$\tilde{\mathbf{x}}_i = \underbrace{\sum_{j \neq i} g_{\mathbf{y}_i}(\mathbf{y}_{i,j})}_{\text{context-mixing}} + \underbrace{g_{\mathbf{y}_i}(\mathbf{y}_{i,i})}_{\text{self-preserving}} + \boldsymbol{\beta}, \quad (19)$$

where $\mathbf{y}_{i,j}$ is given by (8) for $j \neq i$ and by (9) for $j = i$, and $g_{\mathbf{y}_i}$ is defined in (17).

The *context-mixing* terms capture information flowing from other positions into the output at position i, while the *self-preserving* term captures how much of the original input $\mathbf{x}_i$ is retained. In the self-preserving term, information from $\mathbf{x}_i$ flows through two pathways: self-attention (weighted by $\alpha_{i,i}^{(h)}$) and the residual connection.

1.6 Context-Mixing Ratio

Based on the decomposition in (19), we define the context-mixing and self-preserving vectors as:

$$\tilde{\mathbf{x}}_{i\leftarrow\text{context}} := \sum_{j \neq i} g_{\mathbf{y}_i}(\mathbf{y}_{i,j}), \quad (20)$$

$$\tilde{\mathbf{x}}_{i\leftarrow i} := g_{\mathbf{y}_i}(\mathbf{y}_{i,i}). \quad (21)$$

The **context-mixing ratio** is defined as:

$$r_i = \frac{\|\tilde{\mathbf{x}}_{i\leftarrow\text{context}}\|}{\|\tilde{\mathbf{x}}_{i\leftarrow\text{context}}\| + \|\tilde{\mathbf{x}}_{i\leftarrow i}\|}. \quad (22)$$

A higher r_i indicates that context-mixing is dominant in computing $\tilde{\mathbf{x}}_i$; a lower r_i indicates that self-preserving is dominant. The bias term $\boldsymbol{\beta}$ affects neither the context-mixing nor the self-preserving effect, and is therefore excluded from the analysis.

Note that r_i is defined as the ratio of the *norm of the vector sum*, not the sum of individual norms. That is, we first aggregate the contribution vectors, then measure the magnitude of the aggregated vector.

1.7 Stage-Wise Mixing Ratios

To isolate the role of each component in the attention block, we compute mixing ratios at three stages, following the analysis framework of **?**].

1.7.1 Attention Only (Attn-N). Considering only the multi-head attention output (6), without residual connection or layer normalization:

$$\mathbf{c}_i^{\mathrm{Attn}} := \sum_{j \neq i} \sum_{h=1}^{H} \alpha_{i,j}^{(h)} f^{(h)}(\mathbf{x}_j), \quad (23)$$

$$\mathbf{p}_i^{\mathrm{Attn}} := \sum_{h=1}^{H} \alpha_{i,i}^{(h)} f^{(h)}(\mathbf{x}_i), \quad (24)$$

$$r_i^{\mathrm{Attn}} := \frac{\|\mathbf{c}_i^{\mathrm{Attn}}\|}{\|\mathbf{c}_i^{\mathrm{Attn}}\| + \|\mathbf{p}_i^{\mathrm{Attn}}\|}. \quad (25)$$

1.7.2 Attention + Residual (AttnRes-N). Incorporating the residual connection (8)–(9), but without layer normalization:

$$\mathbf{c}_i^{\mathrm{AttnRes}} := \sum_{j \neq i} \sum_{h=1}^{H} \alpha_{i,j}^{(h)} f^{(h)}(\mathbf{x}_j), \quad (26)$$

$$\mathbf{p}_i^{\mathrm{AttnRes}} := \sum_{h=1}^{H} \alpha_{i,i}^{(h)} f^{(h)}(\mathbf{x}_i) + \mathbf{x}_i, \quad (27)$$

$$r_i^{\mathrm{AttnRes}} := \frac{\|\mathbf{c}_i^{\mathrm{AttnRes}}\|}{\|\mathbf{c}_i^{\mathrm{AttnRes}}\| + \|\mathbf{p}_i^{\mathrm{AttnRes}}\|}. \quad (28)$$

Note that $\mathbf{c}_i^{\mathrm{AttnRes}} = \mathbf{c}_i^{\mathrm{Attn}}$, since the residual connection adds $\mathbf{x}_i$ only to the self-preserving term. The residual connection thus increases $\|\mathbf{p}_i^{\mathrm{AttnRes}}\|$ relative to $\|\mathbf{p}_i^{\mathrm{Attn}}\|$, which in general decreases the mixing ratio.

1.7.3 Full Attention Block (AttnResLN-N). Incorporating all three components, using the decomposition derived in (19):

$$\mathbf{c}_i^{\text{AttnResLN}} := \sum_{j \neq i} g_{\mathbf{y}_i}(\mathbf{y}_{i,j}), \tag{29}$$

$$\mathbf{p}_i^{\text{AttnResLN}} := g_{\mathbf{y}_i}(\mathbf{y}_{i,i}), \tag{30}$$

$$r_i^{\text{AttnResLN}} := \frac{\|\mathbf{c}_i^{\text{AttnResLN}}\|}{\|\mathbf{c}_i^{\text{AttnResLN}}\| + \|\mathbf{p}_i^{\text{AttnResLN}}\|}. \tag{31}$$

Kobayashi et al. [1] showed that layer normalization further reduces the context-mixing ratio by scaling (via $\boldsymbol{\gamma}$) in a way that shrinks the attention-derived vectors relative to the residual.

References

[1] Goro Kobayashi, Tatsuki Kuribayashi, Sho Yokoi, and Kentaro Inui. 2021. Incorporating residual and normalization layers into analysis of masked language models. *arXiv preprint arXiv:2109.07152* (2021).